A defining polynomial in twenty-five variables: combined scalar and comparison conditions

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Abstract

Under the four arithmetic inputs of the specified elliptic source, we construct a computable family of integer polynomials in twenty-five total variables, each of degree less than 2^{87}. Every sufficiently large odd value of one fixed coefficient parameter gives a Diophantine definition of the integers over the rationals. The construction places the scalar divisibility conditions and the comparison conditions in two common pole predicates. Rational normalization controls arbitrarily large poles of the conditions that are inactive at a given prime. At primes common to the two arithmetic coefficients, a valuation comparison supplies either the desired integer residual or the scalar index; no prime of the comparison radical is discarded. Two disjoint sets of translated contacts make the scalar conditions available at every required prime. The aggregate trace-height theorem has an ineffective additive constant, so the sufficient coefficient parameter is not numerically certified. Exact source specifications and independent uniform degree checks accompany the existence theorem.

1 Statement and arithmetic inputs

All variable counts include the distinguished rational input. An integer polynomial $F(X, \mathbf{Y})$ defines $\mathbb{Z}$ over $\mathbb{Q}$ if

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \mathbf{y} \in \mathbb{Q}^{\dim \mathbf{Y}} \quad F(x, \mathbf{y}) = 0 \qquad (x \in \mathbb{Q}).$$

The integer L in the following theorem is a fixed coefficient parameter, not a quantified variable.

Theorem 1.1. *Assume the four arithmetic inputs in Section 1.1. There is a computable family of finite polynomial specifications*

$$F_L \in \mathbb{Z}[X, Y_1, \ldots, Y_{24}], \qquad L \geq 3 \text{ odd},$$

with the following properties.

1. *Every member has degree at most*

$$10426327590919878727119667 2 < 2^{87}.$$

   Its number of nonzero monomials is less than 2^{2035}.

2. *There is a fixed* L_0 *such that every odd* $L \geq L_0$ *defines* $\mathbb{Z}$ *over* $\mathbb{Q}$.

3. *The proof does not give an effective sufficient value of* L_0 *or a coefficient-height bound for a member certified to define* $\mathbb{Z}$.

No additional abc hypothesis is used.

The specification uses finite integer polynomial operations, determinants, and finite products with explicit index ranges. Its matrix entries and finite-algebra generators do not count as quantified variables.

1.1 The inherited arithmetic inputs

Put

$$K = \mathbb{Q}(\sqrt{2}), \qquad d = 3 - \sqrt{2}, \qquad D = d^2 = 11 - 6\sqrt{2}, \qquad E : y^2 = x^3 - Dx,$$

and

$$P = \left(\frac{57 - 28\sqrt{2}}{8}, \frac{-660 + 417\sqrt{2}}{32} \right).$$

We use the following inputs, with the scope established in [3].

1. $2E(K) = \mathbb{Z}P$, and $h(x([n]P)) \geq n^2 - 6$ for every nonzero integer n.

2. The modular realization input for the five contacts is the one specified in [4]. Its proved contact-height consequence [7] is as follows. If $q = u/v \in \mathbb{Q}$, written in lowest terms with $v > 0$, is not one of $0, 1, 2401, -343/32$, let $M_5(q)$ be the product of primes $p > 76176$ at which at least one of

   $$v_p(v),\ v_p(u),\ v_p(u - v),\ v_p(u - 2401v),\ v_p(32u + 343v)$$

   is positive and odd, then

   $$h(q) \leq C_3 M_5(q)^3 (1 + \log M_5(q)), \qquad C_3 < 2^{330670}.$$

   In particular, since $1 + \log M \leq 6M^{1/6}$ for $M \geq 1$, we use

   $$h(q) \leq H M_5(q)^{19/6}, \qquad H = 6C_3. \tag{1}$$

   The modular realization is the one inherited from the pinned source and is an explicit hypothesis of this theorem.

3. With $B = 17b$, the pole predicate consists of

   $$p_0^2 = B(1 - e)(1 + 3e), \qquad Bq_0^2 = (s^2 - Be(B - b^2))(s^2 + 3Be(B - b^2)). \tag{2}$$

   A rational solution excludes negative odd valuations of b outside $\{2, 3, 17\}$. A solution exists whenever the reduced denominator of b is a square and

   $$0 < b < 17, \quad b \in \mathbb{Q}_2^{\times 2} \cap \mathbb{Q}_3^{\times 2} \cap \mathbb{Q}_{17}^{\times 2}, \quad v_2(b) \geq 3, \quad v_3(b) \geq 1, \quad v_{17}(b) \geq 2. \tag{3}$$

   We retain the restricted-converse, descent, and simultaneous-realization inputs supporting these assertions.

4. The simultaneous prime-pattern input supplies the prescribed prime values and Gaussian generators, their conjugate reduction-order conditions, arbitrary fixed denominator divisibility, and arbitrary fixed odd avoidance. It applies to any fixed finite list of rational contact coefficients after the root points have been chosen.

The last input is the consequence of Sun's Theorem 4.1 stated and proved in the arithmetic dependency paper. In particular, its denominator divisibility and avoidance requirements may be enlarged by any fixed finite set of primes before it is applied.

The aggregate trace-height theorem [2], obtained from von Buhren's theorem [14], supplies fixed constants $C > 0$ and $J \geq 0$ such that

$$\max_i |n_i|^2 \leq Ch \left(\mathrm{Tr}_{K/\mathbb{Q}} \sum_{i=1}^{4} x([n_i]P) \right) + J \qquad (n_i \in \mathbb{Z} \setminus \{0\}). \tag{4}$$

The additive constant is ineffective. The companion proof verifies that the graph of this fixed trace function has no constant abelian translate through the relevant tuples; this verification includes translates over the algebraic closure. Thus the constants in (4) are uniform in all four labels.

2 The rational source

All displayed rational expressions are restricted to their domains. We require every denominator to be nonzero, every contact value and pole input to be nonzero, and explicitly require $\lambda \neq 0$. Clearing a rational expression always retains its denominator guard.

2.1 Nine points and their finite charts

The nine points are

$$A, S, T_1, Z, H_0, U_A, U_S, U_T, U_Z \in E(K).$$

For each point retain two rational coordinates h_0, h_1, and use

$$h = h_0 + h_1\sqrt{2}, \qquad x = h^2, \qquad j^2 = h^4 - D, \qquad y = hj. \tag{5}$$

These are exactly the nonzero multiples of P. Here is the descent argument. The point $R_0 = (1+\sqrt{2}, 2\sqrt{2})$ satisfies $2R_0 = P$. The group input writes every point of $E(K)$ as $mR_0 + T$ with $T \in E[2](K)$. Under the two-isogeny descent map, $\delta(R_0) = 1+\sqrt{2}$ has opposite signs at the two real embeddings, whereas the torsion classes $1, -D, d, -d$ have equal signs. A square abscissa therefore forces m even. Among these torsion classes only 1 is a square: the negative classes are excluded by signs and $N_{K/\mathbb{Q}}(d) = 7$ excludes d. Conversely every double has square abscissa. Neither $h = 0$ nor $j = 0$ occurs in (5); see [11] for the descent-map proof.

For completeness we specify the single finite chart for j. Write

$$h^4 - D = A_0 + B_0\sqrt{2},$$

where

$$A_0 = h_0^4 + 12h_0^2h_1^2 + 4h_1^4 - 11, \qquad B_0 = 4h_0h_1(h_0^2 + 2h_1^2) + 6.$$

Set $c_0 = 2A_0 + B_0$, $d_0 = -A_0 - B_0$ and

$$A' = 6A_0^3 - 9A_0B_0^2 - 4B_0^3, \qquad B' = -4A_0^3 + 6A_0B_0^2 + 3B_0^3.$$

Adjoin one quartic root Y and reconstruct

$$Y^4 - 4A'Y^2 + 8(B')^2 = 0, \qquad j = \frac{Y/2 + (B'/Y)\sqrt{2}}{c_0 + d_0\sqrt{2}}. \tag{6}$$

If $j = u + v\sqrt{2}$, the rational component of $(c_0 + d_0\sqrt{2})j$ is $2(u^3 - 2v^3)$, which cannot vanish. The norm of its multiplier is $2A_0^2 - B_0^2 \neq 0$. The remaining nonvanishing assertions follow from

$$A'^2 - 2B'^2 = (2A_0^2 - B_0^2)^2(A_0^2 - 2B_0^2),$$

and the discriminant

$$2^{15}B'^2(2A_0^2 - B_0^2)^4(A_0^2 - 2B_0^2)^2 \neq 0.$$

Indeed $h^4 = D$ would imply $N(h)^4 = 49$, and $B' = 0$ would give a rational root of the Eisenstein polynomial $4T^3 - 6T - 3$. Thus the point quartic is separable at every rational retained-base specialization, and its automatic nonzero factors may be omitted from the guard. These statements are proved in [10, 12].

Write the integer labels of (A, S, T_1, Z) as (n, a, t, b), and put

$$c = |a - 1|, \qquad B_{\mathrm{ind}} = \max(|n|, |a|, |t|, |b|).$$

The affine formula for $S - P$ excludes $S = \pm P$, so $c > 0$. Set $Q_0 = P$, $(Q_1, Q_2, Q_3, Q_4) = (A, S, T_1, Z)$, $U_0 = H_0$, and $(U_1, U_2, U_3, U_4) = (U_A, U_S, U_T, U_Z)$. Define $V_i = U_i - Q_i$ for $0 \leq i \leq 4$. These differences use the usual affine group formulas and their nonzero equal-abscissa guards.

2.2 Comparison and scalar functions

Write $f(Q) = 1/x(Q)$ and $z(Q) = -x(Q)/y(Q)$. To avoid confusion with this local parameter, the scalar coefficient below is denoted ρ. The fourth-order formal-logarithm truncation is simply

$$\ell_4(Q) = z(Q).$$

Put

$$L_i = \frac{\ell_4(U_i)}{\ell_4(H_0)}, \qquad L_i' = \sigma\left(\frac{\ell_4(V_i)}{\ell_4(V_0)}\right) \quad (1 \le i \le 4),$$

where σ is the nonidentity automorphism of K. Let

$$\mathcal{F}(u,v,w) = v^2 - u(u-2)vw + (u-2)^2w^2 - 1, \qquad \beta = 216X^4 + 450459.$$

The sixteen comparison functions $C_1, \dots, C_{16}$, in order, are

$$\begin{array}{cccc} & (f(U_i)^2)_{i=0}^4, & ((\sigma f(V_i))^2)_{i=0}^4, & \\ L_i - L_i' \quad (1 \le i \le 4), & L_1 - \beta, & \mathcal{F}(L_2, L_3, L_4). & \end{array} \tag{7}$$

The first line contains ten functions. The exponent 2 permits high valuation of these functions while requiring only first formal neighborhoods for the underlying points.

The six scalar functions $S_1, \dots, S_6 \in \mathbb{Q}$ are the two rational coefficients, in the basis $1, \sqrt{2}$, of each of

$$f(H_0), \qquad f(S-P), \qquad z(S-P)/z(H_0), \tag{8}$$

in that order. In particular a lower valuation bound on all six functions holds at every place of K above the given rational prime.

2.3 The scalar coefficient and two contact functions

For fixed odd $L \ge 3$, define

$$\begin{gathered} \ell = 76249, \qquad r = \beta(\beta-1)(\beta-2401)(32\beta+343)(2\beta-1), \\ C_L = 17(\ell^{(L+1)/2} + 1), \qquad \rho = \frac{102^2 \ell^L r}{(1 + C_L r)^2}. \end{gathered} \tag{9}$$

The normalization theorem and its finite residue certificate [8] prove

$$v_\ell(\rho) = \begin{cases} L, & v_\ell(X) \ge 0, \\ L - 20v_\ell(X), & v_\ell(X) < 0. \end{cases} \tag{10}$$

This valuation is always positive and odd. For integral X the reduced denominator of ρ is a square, ρ is a square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$, and its valuations there satisfy the preparation bounds in (3). In fact $v_2(\rho) = 4$. For every real X,

$$0 < \rho < \frac{36}{\ell r} < 1. \tag{11}$$

Let

$$\Theta = \{0, -1/2, -1, 1/2399, -32/407\},$$

$$\Theta_0 = (2+\Theta) \cup (4+\Theta), \qquad \Theta_1 = (6+\Theta) \cup (8+\Theta), \qquad \gamma_0 = -1, \quad \gamma_1 = -2.$$

The two contact-and-pole sets are disjoint and contain no zero. Define

$$T = \mathrm{Tr}_{K/\mathbb{Q}} \sum_{i=1}^{4} x(Q_i), \qquad A_0 = 1, \qquad A_1 = (1 + \rho T^2)^2, \qquad t_j = A_j \lambda, \tag{12}$$

and

$$\delta_j = t_j - \gamma_j, \qquad R_j = \frac{\prod_{\theta\in\Theta_j}(t_j-\theta)}{\delta_j^{10}} \quad (j=0,1). \tag{13}$$

All abscissas in (12) are positive at both real embeddings, so $T>0$ and $A_1>0$.

Let $\mathcal{S}$ consist of all primes at most 76176, together with the prime divisors of the numerators and denominators of the 22 contact-and-pole values and of their nonzero pairwise differences. Exact rational arithmetic gives

$$\mathcal{S}\setminus\{p : p\le 76176\} = \{1875611, 3828397, 3982747\}. \tag{14}$$

In particular $\ell\notin\mathcal{S}$.

2.4 Two pole inputs

For any nonzero rational coefficient q set

$$\Psi_q(v) = \frac{qv^2}{1+qv^2}.$$

For $j=0,1$, put $\mathcal{R}_j=\rho R_j$ and

$$E_i = \rho^7\Psi_\rho(C_i), \qquad G_{j,k} = R_j^7\Psi_{R_j}(S_k). \tag{15}$$

Define

$$W_j = \prod_{i=1}^{16}\left(1+\mathcal{R}_j\left(\frac{E_i}{\delta_j\mathcal{R}_j^7}\right)^2\right)\prod_{k=1}^{6}\left(1+\mathcal{R}_j\left(\frac{G_{j,k}}{\mathcal{R}_j^7}\right)^2\right), \tag{16}$$

and

$$b_j = \frac{4\rho R_j}{\mathrm{Tr}_{K/\mathbb{Q}}(W_j^{-1})^2}. \tag{17}$$

Both inputs receive the pole predicate (2); both scalar and comparison functions occur in these two inputs.

Each first conic is represented by

$$e = \frac{\tau^2-B}{\tau^2+3B}, \qquad p_0 = \frac{-4B\tau}{\tau^2+3B}, \qquad B = 17b_j,$$

with retained coordinates τ, s. The second equation is a single quadratic equation for q_0, guarded nonzero. The inherited realization provides $p_0q_0s\neq 0$ and $e\neq 1$ under (3), as proved in [9]. Indeed its two elliptic points have nonzero abscissas of the prescribed squareclass and are outside two-torsion, so both ordinates are nonzero. Rescaling gives the nonzero witnesses in (2); then $\tau = p_0/(e-1)$ gives the displayed chart. This restriction changes neither the inherited obstruction nor its required converse.

3 Local extraction at every required prime

At primes outside $\mathcal{S}$, valuations on completions of K are normalized by $v(p)=1$ and have integer values. All arguments below permit zero test functions, whose valuations are $+\infty$.

Lemma 3.1. *If $v(q)$ is odd, then*

$$v(\Psi_q(v_0)) = \max\{0, v(q)+2v(v_0)\}.$$

For arbitrary nonzero q,

$$v(\Psi_q(v_0)) \le \max\{0, v(q)+2v(v_0)\}.$$

If q is a unit, positive valuation of $\Psi_q(v_0)$ is equivalent to positive valuation of v_0, and then equals $2v(v_0)$.

Proof. Write $e = v(q) + 2v(v_0)$. If $e \neq 0$, subtraction of $v(1 + qv_0^2) = \min(0, e)$ gives the first formula. If $e = 0$, the numerator is a unit and the denominator is integral, so the quotient has nonpositive valuation. This also proves the general upper bound and the unit assertion. Odd $v(q)$ excludes $e = 0$. □

Lemma 3.2. *Suppose $v_p(\rho R_j) = \alpha$ is odd and b_j has a pole certificate. Then $\alpha > 0$, and at some place above p each factor of W_j has valuation greater than $-\alpha/2$. Every scalar numerator $G_{j,k}$ at that place has valuation greater than $25\alpha/4$. If $v_p(\delta_j) = 0$, the same holds for each comparison numerator E_i.*

Proof. Every nonzero second summand in (16) has odd valuation. Thus every factor, and hence W_j, has nonpositive valuation. Set $q = \max_{\mathfrak{p}|p} v_{\mathfrak{p}}(W_j) \leq 0$. Integrality of local traces gives $v_p(\mathrm{Tr}(W_j^{-1})) \geq -q$. Therefore

$$1 \leq v_p(b_j) \leq \alpha + 2q,$$

because $v_p(b_j)$ is odd and the pole predicate excludes negative odd values. In particular $\alpha > 0$ and $q > -\alpha/2$. At a place attaining q, each nonpositive factor valuation is at least their sum q. A scalar numerator V, or a comparison numerator with unit δ_j, consequently satisfies

$$2v(V) - 13\alpha > -\alpha/2, \qquad v(V) > \frac{25}{4}\alpha.$$

□

Lemma 3.3. *If $p \notin \mathcal{S}$ and $\mu = v_p(\rho)$ is odd, at least one R_j is a unit. A source solution then has $\mu > 0$ and*

$$v_p(S_k) > \frac{25}{8}\mu \quad (1 \leq k \leq 6), \qquad v_p(c) > \frac{25}{8}\mu.$$

In particular $v_p(c) \geq 4$ and $v_p(c) \geq 3\mu + 1$.

Proof. Since $v_p(\rho T^2)$ is odd, either $A_1 \equiv 1 \pmod{p}$ or $v_p(A_1) < 0$. For either contact function, an argument of positive or negative valuation gives a unit: at negative valuation numerator and denominator have the same degree, while at positive valuation all contact and pole constants are units. If λ is a unit and R_0 is not, its residue is a contact or pole of the first set. Either $A_1\lambda$ has negative valuation, or it has the same residue as λ. Disjointness of the two reduced sets gives a unit R_1 in both cases. This is the unit assertion of [6].

At the chosen detector, $v_p(\delta_j) \leq 0$. Lemma 3.2 excludes $\mu < 0$. By Lemma 3.1, $v(E_i) \geq 7\mu$; thus all comparison factors in (16) are units, even when the original comparison functions have poles. The scalar numerators satisfy $v(G_{j,k}) > 25\mu/4$ at the place supplied by the trace lemma. Since R_j is a unit, this implies $v_p(S_k) > 25\mu/8$.

The first four scalar components put H_0 and $S - P$ in the formal group at every place over p. Write $H_0 = [h]P$; the point guard gives $h \neq 0$. The formal logarithm is an isometry and $[h](S - P) = [a - 1]H_0$, so

$$v_{\mathfrak{p}}(z(S - P)/z(H_0)) = v_p((a - 1)/h).$$

The last two rational components give a lower bound $> 25\mu/8$ on the left. Since h is an integer, the asserted bound for c follows. □

Lemma 3.4. *Let $p \notin \mathcal{S}$ and $\nu = v_p(R_j) > 0$ be odd. A source solution has one of the following two properties:*

1. *at some place over p, every C_i has valuation at least 4;*

2. $v_p(c) \geq 4$.

Proof. Here δ_j is a unit. Put $\mu = v_p(\rho)$. If μ is odd, Lemma 3.3 proves the second alternative, or excludes the solution when $\mu < 0$.

Suppose μ is even. Lemma 3.2 gives $\alpha = \mu + \nu > 0$ and one place where both numerator families have valuation $> 25(\mu+\nu)/4$. For a nonzero comparison function with $u = v(C_i)$, Lemma 3.1 gives

$$v(E_i) \leq 7\mu + \max(0, \mu + 2u).$$

If $0 \leq \mu \leq \nu$, it follows that

$$u > \frac{25\nu - 7\mu}{8} \geq \frac{9}{4}\nu.$$

For $\nu = 1$, evenness forces $\mu = 0$, so $u > 25/8$ and $u \geq 4$. For $\nu \geq 3$, the lower bound $u > 9\nu/4$ is stronger. If $\mu < 0$, the same calculation gives $u > (25\nu - 7\mu)/8 > 25\nu/8 \geq 25/8$, again sufficient. The possible cancellation when $\mu + 2u = 0$ is covered by the upper bound and cannot satisfy the strict required inequality.

It remains that $\mu > \nu > 0$. For $s = v_p(S_k)$ the oddness of ν gives

$$v(G_{j,k}) = 7\nu + \max(0, \nu + 2s).$$

Consequently

$$s > \frac{25\mu - 7\nu}{8} \geq \frac{43}{8}.$$

The last lower bound uses $\nu \geq 1$, $\mu \geq \nu + 1$, with μ even and ν odd. All six rational scalar components therefore have valuation at least 6. The formal-logarithm argument in Lemma 3.3 then gives $v_p(c) \geq 6 \geq 4$. This proves the second alternative and completes all valuation cases. □

3.1 From comparison functions to integer residuals

The formal-logarithm estimate needed here has a direct short proof. If Q is in the first formal neighborhood at a prime $p > 76176$ and $r_Q = v(z(Q)) \geq 1$, then

$$v(\log_E(Q) - \ell_4(Q)) \geq 5r_Q, \qquad \ell_4(Q)/\log_E(Q) \in 1 + p^4\mathcal{O}. \tag{18}$$

Indeed the full logarithm is $z(Q)\sum_{j\geq 0} a_j (Df(Q)^2)^j$, with $a_0 = 1$, $a_1 = -2/5$ and $a_j = a_{j-1}(4j-5)/(4j+1)$ for $j \geq 2$. One has $v(Df(Q)^2) = 4r_Q$, and the omitted terms start at index 1. The denominator of a_j divides $(4j+1)!$. Terms with $4j+1 < p$ have integral coefficients; for the remaining terms, $j \geq 19044$ and $v(a_j(Df^2)^j) \geq 4jr_Q - j \geq 3jr_Q \geq 4r_Q$. These bounds tend to infinity and prove (18).

Suppose the first alternative of Lemma 3.4 holds. The ten powered functions in (7) put H_0, U_i in the formal group at the selected place and V_0, V_i in the formal group at its conjugate. Write $U_i = [u_i]P$ and $H_0 = [h]P$. The full logarithm ratios are

$$r_i = u_i/h, \qquad r_i' = (u_i - n_i)/(h-1), \qquad (n_1, n_2, n_3, n_4) = (n, a, t, b).$$

At least one of $h, h-1$ is a p-unit. Its ratio is integral; the cross test and (18) make the other truncated ratio integral as well. Both relative errors are then absolute errors. Modulo p^4 one has $L_i \equiv r_i \equiv r_i'$. Multiplication by $h, h-1$ and subtraction give

$$L_i \equiv n_i \pmod{p^4}. \tag{19}$$

This argument is valid at split and inert primes.

Define the integer residual

$$\Delta = \mathcal{F}(a, t, b).$$

Then $p^4 \mid \Delta$. If $\beta = u/v$ is in lowest terms with $v > 0$, the direct target and (19) make β integral at p and give $p^4 \mid u - nv$.

Let M_j be the product of primes outside $\mathcal{S}$ at which R_j has positive odd valuation. For either nonzero integer $N = \Delta$ or $N = u - nv$, Lemma 3.4 therefore gives

$$M_j^4 \mid Nc \qquad (j = 0, 1). \tag{20}$$

Primes occurring in both arithmetic coefficients are included in this divisibility statement.

4 Heights and soundness

4.1 The scalar height and fixed-prime lower bound

Every prime contributing to $M_5(\beta)$ outside $\mathcal{S}$ has positive odd valuation in ρ. To check this, a negative valuation of $\beta = 216X^4 + 450459$ at such a prime is a multiple of four and cannot contribute an odd denominator contact. At a finite contact exactly one factor of r has positive odd valuation; the denominator $1 + C_L r$ is a unit. The prime ℓ contributes no contact, by the finite residue certificate for (10). Lemma 3.3 thus gives, with fixed H_1,

$$h(\beta) \le H_1 c^{19/24}. \tag{21}$$

Omitting the three additional primes in (14) only multiplies the height constant by their product to power 19/6. At the fixed prime ℓ, the same lemma and (10) give

$$v_\ell(c) \ge 3L + 1, \qquad c \ge \ell^{3L+1}, \qquad L \log \ell \le \frac{\log c}{3}. \tag{22}$$

The scalar coefficient has the elementary height bound

$$h(\rho) \le 15h(\beta) + 2L \log \ell + C_0, \tag{23}$$

where C_0 is independent of X, L. In fact $h(r) \le 5h(\beta) + \log(2 \cdot 2402 \cdot 375 \cdot 3)$, and

$$h(\rho) \le \log(102^2) + L \log \ell + 3h(r) + 2 \log C_L + 2 \log 2.$$

The bound $C_L \le 34\ell^{(L+1)/2}$ proves (23).

4.2 Height from two disjoint groups of contacts

For $s \in \{2, 4, 6, 8\}$ the map

$$q_s = 2 + \frac{1}{t - s}$$

sends the contacts $s + \Theta$ to $\infty, 0, 1, 2401, -343/32$. Its inverse has fixed rational coefficients. Outside $\mathcal{S}$, the odd contact radical of q_s consists exactly of the primes at which t meets one of these five contacts with odd multiplicity. Within each R_j the two groups are disjoint modulo every such prime. Consequently their radicals multiply to M_j, and applying the contact-height theorem to each group and multiplying gives a fixed constant C_1 with

$$h(t_j) \le C_1 M_j^{19/12}. \tag{24}$$

For clarity, each separate application gives $h(t_j) \le C_s M_s^{19/6}$ with a fixed C_s; their product gives the square of (24). Fixed exceptional primes are absorbed into C_s.

The nonzero slope gives $A_1 = t_1/t_0$. Hence

$$h(A_1) \le h(t_0) + h(t_1), \qquad h(A_1) = 2h(1+\rho T^2) \ge 4h(T) - 2h(\rho) - 2\log 2.$$

Combining (4), (20), (23), and (22) yields fixed constants $C_2 \ge 1$, $J_2 \ge 0$, independent of X, L, such that

$$B_{\mathrm{ind}}^2 \le C_2|Nc|^{19/48} + C_2 c^{19/24} + C_2 \log c + J_2 \tag{25}$$

for each nonzero residual N under consideration.

4.3 Eliminating the residuals

Write $B = B_{\mathrm{ind}}$ in this subsection. From (22), $B \ge c - 1 > 4$ and $c \le B + 1 \le 2B$. If $\Delta \ne 0$, then $|\Delta| \le 5B^4$, so (25) implies

$$B^2 \le C_2 10^{19/48} B^{95/48} + C_2 (2B)^{19/24} + C_2 \log(2B) + J_2.$$

Every term on the right grows more slowly than B^2. There is therefore a fixed threshold, independent of L, beyond which this inequality is impossible. Sufficiently large c forces $\Delta = 0$.

The integer Lucas growth lemma of the source [9] then gives

$$\log B > c - 4. \tag{26}$$

If $\beta \ne n$, let $N = u - nv \ne 0$ with $\beta = u/v$ in lowest terms. Since $\max(|u|, v) = e^{h(\beta)}$,

$$|Nc| \le 4e^{h(\beta)} B^2.$$

For sufficiently large c, the last three terms on the right of (25) are at most $B^2/2$, because $B \ge c - 1$. Putting $\eta = 19/48$ gives

$$B^{2-2\eta} \le 2C_2 4^{\eta} e^{\eta h(\beta)}, \qquad \log B \le \frac{\log(2C_2 4^{\eta}) + \eta H_1 c^{19/24}}{2 - 2\eta}.$$

This contradicts (26) for sufficiently large c. Thus $\beta = n \in \mathbb{Z}$.

For rational X, integrality of $216X^4 + 450459$ is equivalent to $X \in \mathbb{Z}$: in a reduced fraction $X = u/v$ it implies $v^4 \mid 216 = 2^3 3^3$, and hence $v = 1$. All sufficient lower bounds on c above are fixed independently of L. Equation (22) therefore proves soundness for every sufficiently large fixed odd L. The only ineffective constant used here is inherited from (4).

5 Completeness

Fix $X \in \mathbb{Z}$ and any fixed odd $L \ge 3$. We prove completeness for this value of L, without a lower-bound hypothesis on L. The order of choices below is needed because the prime-pattern theorem is applied only after all roots, scales, local neighborhoods, and bounds are fixed.

5.1 Preparing the scalar functions

Let $\mathcal{P}$ be the union of the finite prime support of ρ, the exceptional set $\mathcal{S}$, and the primes $2, 3, 17$. For the shared point $H_0 = [h_0]P$, choose a positive h_0 divisible by 64 so that H_0 belongs to sufficiently deep formal subgroups at these finite places and is close to the identity at both real embeddings. After imposing the finite divisibility, such real returns are possible in the compact closure of the cyclic subgroup. They do not require this closure to be the full product of the two real elliptic groups.

Next choose a positive c divisible by 6 and by any further required finite modulus, with $[c]P$ still closer to the identity at all these places. Set $a = c + 1$ and $S = [a]P$. This makes the

six rational components in (8) arbitrarily small simultaneously in the real absolute value and at the specified finite primes. The ratio is controlled because H_0 has already been fixed and has nonzero local parameter. The finite-place assertion is exactly the scalar realization in the source, including ramified and bad primes; its proof uses sufficiently deep formal subgroups and continuity [5]. The real assertion follows from the same continuity and positive return times in the compact cyclic closure.

For the real preparation below, choose the scalar components small enough that all six scalar factors in (16) are as close to 1 as required whenever $1/2 < R_j < 2$. At a fixed finite prime, choose them small enough for the same conclusion whenever R_j is sufficiently close to 1. These choices are possible: as $S_k \to 0$, each scalar second summand tends to zero, with ρ fixed. No condition involving the as yet unchosen slope is needed.

Set $n = \beta$. The explicit Lucas construction takes

$$U_0 = 0, \quad U_1 = 1, \quad U_{m+1} = aU_m - U_{m-1}, \quad k = 2(a-2), \quad b = U_k/(a-2), \quad t = U_{k+1}.$$

The congruence $U_m \equiv m \pmod{a-2}$ shows that b is an integer; the Lucas identity gives $\mathcal{F}(a,t,b) = 0$. As a is odd and $a \equiv 1 \pmod 3$, the sequence modulo two has period $0, 1, 1$ and $k \equiv 1 \pmod 3$, so t, b are positive and odd. The four root points and both positive square scales in (12) are now fixed.

Temporarily set $H = [h]P$ and $U_i = [hn_i]P$. Restrict the shared index to a fixed congruence class and a compact real neighborhood on which all prepared scalar bounds persist. The comparison functions and their normalizations are continuous wherever defined. Shrinking the neighborhoods, and avoiding their finitely many poles under the fixed multiplication and translation maps, bounds all comparison functions there. These rational functions have no identically vanishing normalization denominators: $\rho > 0$ makes $1 + \rho C_i^2 > 0$ at each real embedding. The fixed congruence class may be taken modulo a number divisible by 64.

5.2 Choosing the slope and comparison primes

Apply the inherited simultaneous prime-pattern theorem to the rational coefficients $-\theta/A_j$, for both j and all $\theta \in \Theta_j$. Repeated coefficients are listed once, retaining their incidences. In the notation of its rational-contact form, choose

$$\lambda = u/v, \qquad v = D_0^2 w^2,$$

so that all $u - (\theta/A_j)v$ are prescribed positive primes. Require these primes to avoid every previously fixed prime, the numerator and denominator support of the scales, and all contact differences. Prescribe enough divisibility of w that λ has arbitrarily negative valuation at each prime in $\mathcal{P}$. The same theorem supplies an arbitrarily large positive real slope. The rational-contact and avoidance lemmas of [3] prove these assertions from the stated prime-pattern input.

As λ tends to infinity at any of the preparation places, both R_j tend to 1 and δ_j^{-1} tends to zero. The scalar factors are close to 1 by their preparation. The comparison factors tend to 1 because of δ_j^{-1}, uniformly on the already bounded comparison neighborhoods. Thus all required finite and real closeness conditions on W_j can be imposed before the prime-pattern theorem is used.

Its prescribed conjugate reduction orders give a common arithmetic progression for the final index h: the primary orders divide h, the opposite orders divide $h-1$, and h remains in the previously fixed congruence class. Their odd coprimality and exact two-adic compatibility are proved in [3]. Positive indices in this progression meet the chosen real neighborhood by density in the appropriate component of the compact cyclic closure. Choose such an $h > 2$ and set $H_0 = [h]P$, $U_i = [hn_i]P$.

At a selected comparison prime, $v_p(R_j) = 1$ at each incidence, while ρ and δ_j are units. The reduction-order conditions put the relevant U_i, V_i in their first formal neighborhoods. Consequently $v(f) \geq 2$, and the first ten comparison functions have valuation at least 4. The exact full ratios are the integers n_i; by (18), the other six comparison functions also have valuation at least 4. Hence

$$v(E_i) \geq 8, \qquad v\left(\mathcal{R}_j(E_i/(\delta_j \mathcal{R}_j^7))^2\right) \geq 1 + 16 - 14 = 3.$$

All comparison factors are units congruent to 1 at the designated place. The scalar factors are units congruent to 1 at both places, regardless of possible poles of the scalar functions at these new primes, because odd $v_p(R_j) = 1$ gives $v(G_{j,k}) \geq 7$.

5.3 Preparing the pole inputs

The factorization of the contact numerators gives the exact squareclass description

$$R_j = \left(\frac{A_j}{v\delta_j}\right)^{10} \prod_{\theta \in \Theta_j} (u - (\theta/A_j)v).$$

Thus each R_j is a positive rational square times its ten selected primes. Their odd valuations are positive and equal to one, and they are disjoint from the fixed support of ρ. At those primes $W_j^{-1} \equiv 1$ at the designated place. At its conjugate, each factor of W_j has nonpositive valuation: it is either a unit congruent to 1 or has negative valuation, since its second summand has odd valuation. Thus W_j^{-1} is congruent to 1 or 0 there. The trace is consequently congruent to 2 or 1 modulo p and is a unit, so the odd valuation of b_j is positive. At primes in the support of ρ, both R_j and W_j have been made arbitrarily close to 1. Every odd valuation of ρR_j is therefore nonnegative in b_j. At all remaining primes its valuation is even and remains even after division by the square trace. The reduced denominator of each b_j is a square.

At $2, 3, 17$, both R_j and W_j can be as close to 1 as needed. Each R_j is then a local square, and the trace tends to 2. The squareclass of b_j agrees with that of ρ. At two the factor 4 in (17) cancels the valuation of the limiting trace square, giving $v_2(b_j) = 4$. At three and seventeen the required depth is unchanged. Thus all finite conditions in (3) hold.

At each real embedding arrange $1/2 < R_j < 2$ and $1 \leq W_j < 2$. The latter inequality follows by making the finitely many positive second summands sufficiently small. The trace is then at least 1, so

$$0 < b_j < 8\rho < 17.$$

All normalization denominators, products, and traces are nonzero by positivity. The inherited converse supplies the two pole witnesses with the nonzero chart conditions already described. All nine point charts and the guarded differences are available, since the chosen labels are nonzero, $a > 1$, and $h > 2$. This completes the rational source construction for every integral input and every fixed odd $L \geq 3$.

6 A single polynomial and its degree

The retained rational coordinates are

Coordinates	Number
Input	1
Nine point-coordinate pairs	18
Nonzero slope	1
Two pole-parameter pairs	4
Total before finite-root elimination	24

There are nine quartic point roots and two quadratic pole roots. Their finite-algebra rank is $4^9 2^2 = 2^{20}$. Both pole roots are guarded nonzero. The separating-element theorem [1] eliminates all eleven finite roots and retains every nonautomatic guard with one additional rational coordinate. This gives exactly twenty-five total polynomial variables.

We specify that elimination explicitly. Let B_0 be the algebra of the nine point roots, A_0 the algebra after adjoining the two pole roots q_1, q_2, and write $q_j^2 = c_j \in B_0$ after rational square clearing. Put $b_0 = 2^{18}$ and $n_0 = 2^{20}$. The source guard has the form $G = gq_1q_2$, $g \in B_0$, and there are no additional closed residual equations. Set

$$T_0 = g^2 c_1 c_2, \qquad \det_{B_0}(I - vM_{T_0}) = \sum_{j=0}^{b_0} a_j v^j, \qquad a_0 = 1.$$

Let $\theta_1, \ldots, \theta_m$, $m = 11$, be the point and pole generators, and let δ_i be the discriminant of the ith point quartic. These discriminants never vanish at rational base specializations. Define

$$R = n_0 \left(1 + \sum_{i=1}^{m} \sum_{u,v=1}^{n_0} (M_{\theta_i}^{A_0})_{uv}^2 + \sum_{u,v=1}^{b_0} (M_g^{B_0})_{uv}^2 \right).$$

For $1 \le r \le b_0$, put

$$c_r = a_r, \qquad h_r(Z) = \sum_{j=0}^{r} a_j Z^{r-j}, \qquad E_r = c_r - h_r(T_0),$$

$$D_r = c_r^2 \prod_{i=1}^{9} \delta_i^2, \qquad s_r = 3r + 108,$$

$$K_r = (2R)^{s_r}(1 + c_r^2) \prod_{i=1}^{9} (1 + \delta_i^2), \qquad C_r = D_r + 4RK_r,$$

$$\beta_r = D_r E_r \left(C_r^m + \sum_{i=1}^{m} D_r^i C_r^{m-i} \theta_i \right), \qquad H_r(U) = \det_{A_0}(I - UM_{\beta_r}).$$

The required integer polynomial is

$$F_L = H_{b_0}(U) \prod_{r=1}^{b_0 - 1} \left(H_r(U)^2 + \sum_{j=r+1}^{b_0} a_j^2 \right). \tag{27}$$

The only additional rational coordinate is U.

The proof in [1] handles every base specialization. At the actual degree of $\det(I - vM_{T_0})$, the element E_r/c_r is the idempotent selecting exactly the factors where the guard is nonzero. The coefficient c_r and the nonzero quartic discriminants give lower bounds for the separations of the finite roots. The powers of C_r/D_r distinguish the surviving characters, and multiplication by E_r kills all other factors, including their nilpotents. A rational root of the corresponding H_r is therefore equivalent to a guarded rational character of the finite algebra. The squared tail coefficients exclude incorrect degree cases; when $c_r = 0$ one has $H_r = 1$. Thus (27) has exactly the rational projection of the guarded source, including specializations with an empty nonzero part or complex residue factors.

The certified weights of the nine point generators are 6; the two pole-generator weights are $1181594, 6716314$. The full guard and point-guard weights are $9642537, 1744629$. The separating elements have weight at most $W_1 r + W_0$, where

$$W_1 = 1446676194, \qquad W_0 = 35052024698.$$

The principal-minor degree estimate, summed over every factor, is

$$\deg F_L \le n_0(W_1 b_0^2 + (W_0+1)(2b_0-1)) = 104263275909198787271196672 < 2^{87}.$$

The uniform checker treats ℓ^L and $\ell^{(L+1)/2}$ as independent coefficient symbols of weight zero. Consequently this degree bound holds for every fixed odd L. Integer specialization cannot increase it. Equations (22) and (25), followed by completeness, now prove Theorem 1.1.

There is also a uniform bound on the number of nonzero monomials. Let $\mathcal{D} = 104263275909198787271196672$. From (27),

$$\deg_U F_L \le \mathcal{E} = n_0(2b_0-1) = 549754765312.$$

Counting monomials of total degree at most $\mathcal{D}$ in the 25 variables, with the U exponent at most $\mathcal{E}$, gives

$$\#\operatorname{supp}(F_L) \le \binom{\mathcal{D}+25}{25} - \binom{\mathcal{D}-\mathcal{E}+24}{25} < 2^{2035}.$$

The last inequality is checked by exact integer arithmetic. It is a support upper bound, not an assertion of the expanded monomial count.

7 Exact specifications and reproducibility

The directory `computation/few_variables/merged25` contains the source builder, the guarded rational source, its integer polynomial specification, the independent source checker, and the uniform degree certificate. Its finite-root elimination is in the subdirectory `active25`. The source has 7294 rational coefficient nodes, 2565 integer coefficient nodes, and 152 atomic guard factors. The checks include the nonzero slope, all ten squares, all normalization denominators, the two pole charts, the guard factorization, and the generator dependencies. The active specification has 262144 finite-product factors. These are specification counts; the expanded monomial count is not asserted.

The supplied numerical fixture uses $L = 3$. It checks the source and finite-polynomial construction for a concrete coefficient specialization; it is not certified to satisfy the ineffective sufficient threshold. The defining-polynomial conclusion concerns every sufficiently large odd member of the family. All programs for these source and degree checks use the Python standard library.

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